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**In all quizzes and exams, show your work and write with correct notation.**

(10pts) 1. Define reflexive.

Two samples

Let $R$ be a relation on $A$. Then $R$ is reflexive if $\forall z \in A$, $(z,z) \in R$.

The relation $R$ on set $S$ is reflexive if for all $m \in S$, $mRm$.

(10pts) 2. Consider the relation $R = \{(j,k) \in \mathbb{Z} \times \mathbb{Z} : 10 \text{ divides } j-k\}$ on $\mathbb{Z}$. State two elements of $R$.

$(11,1) \in R$ $(73425, 35) \in R$ $(9,9) \in R$

$(123,43) \in R$

$(38,8) \in R$ $(0,0) \in R$

$(2,-8) \in R$

$(8,38) \in R$ $(10,20) \in R$

$(12, -12345678) \in R$

(10pts) 3. Use the hypotheses

- H1: $W \subseteq M$
- H2: $X \subseteq Y$
- H3: For all $u \in M$ for all $w \in S$, if $u$ buys lunch for $w$, then $w$ buys lunch for $u$.

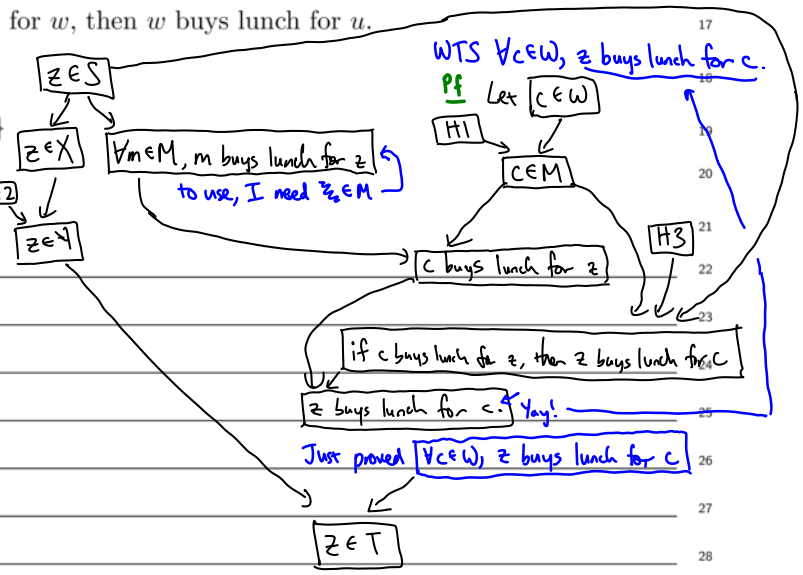
and the definitions

This $\forall$ gets used

- $S = \{b \in X : \text{for all } m \in M,\ m \text{ buys lunch for } b\}$
- $T = \{a \in Y \mid \text{for all } c \in W,\ a \text{ buys lunch for } c\}$

to prove: $S \subseteq T$.

This $\forall$ gets proved




Let $z \in S$. So $z \in X$ and for all $m \in M$, $m$ buys lunch for $z$. Since $z \in X$, by H2, $z \in Y$.

This is a "for all" that I can't use yet. Looking for an element of M.

We want to prove $\forall c \in W$, $z$ buys lunch for $c$. Let $c \in W$. WTS $z$ buys lunch for $c$. Since $c \in W$, by H1, $c \in M$. Using the "for all" in the previous paragraph, $c$ buys lunch for $z$. By H3, we get: $c$ buys lunch for $z$ implies $z$ buys lunch for $c$. Applying modus ponens, $z$ buys lunch for $c$.

The previous paragraph proved $\forall c \in W$, $z$ buys lunch for $c$. This together with $z \in Y$ from the first paragraph lets us conclude $z \in T$. ∎